

The Data Logger is a high-performance LTE Cat 1 OpenCPU IoT development board powered by the Quectel EC200U module. Designed for industrial-grade and remote applications, it combines powerful processing, integrated LTE and GPS antennas, and a wide range of communication interfaces in a compact form factor. It supports QuecPython, enabling direct Python-based development on the module without an external microcontroller – reducing complexity, cost, and power consumption. With advanced MPPT-based solar charging, dual SIM support, and multi-protocol connectivity (UART, I²C, SPI, RS232, RS485), the board delivers exceptional flexibility for scalable IoT, telemetry, and data logging systems.

Key Features

- OpenCPU architecture based on Quectel EC200U LTE Cat 1 module
- Built-in LTE and GPS antennas for cloud connectivity and location tracking
- Dual SIM switching for reliable network redundancy
- MPPT solar charging and power management with battery and USB-C input
- Multi-protocol interfaces: UART, I²C, SPI, RS232, and RS485
- Integrated RTC and data logging for time-stamped sensor monitoring
- Supports QuecPython (MicroPython) for rapid IoT application development
- Compact, rugged design suitable for field deployment



Fig: Data Logger

Technical Specifications

Parameter	Value
Processor Module	Quectel EC200U LTE Cat 1
Input Voltage Range	5 V – 12 V DC
Operating Temperature	–30 °C to +75 °C
Storage Support	microSD card slot (up to 32 GB)
UART Interfaces	3× UART (Main, Debug, Auxiliary)
I ² C / SPI	2× I ² C, 1× SPI
Industrial Interfaces	RS232 and RS485 industrial interfaces
Wireless Connectivity	LTE (4G), GPS
Power Management	MPPT solar charging via CN3791, Li-ion battery, USB-C, and terminal input
Protection Circuit	SMAJ18A TVS diodes for surge and ESD protection
Real-Time Clock	RV-3028-C7 with backup power

Applications

- Smart Agriculture – Real-time monitoring of soil, weather, and irrigation systems using cloud connectivity and solar power
- Industrial Automation – Remote equipment control, predictive maintenance, and data logging over LTE and RS485 networks
- Environmental Monitoring – Collection of air quality, temperature, and humidity data using onboard sensors and GPS tracking
- Asset Tracking – Location-based monitoring of mobile assets with built-in LTE and GPS antennas for reliable data transmission
- Remote Data Logging – Continuous acquisition and cloud upload of multi-sensor data for smart city and IoT applications
- Weather Stations – Integration with sensors for rainfall, temperature, and humidity logging with real-time reporting
- Smart Infrastructure – Used in connected streetlights, water management, and energy metering systems
- IoT Gateways – Acts as a central hub for sensor nodes, aggregating and transmitting data to cloud platforms securely